

Yahya Masri

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EDUCATION

George Mason University

May 2025 - May 2027

Bachelor of Science, Computer Science

- **GPA:** 4.0
- **Coursework:** Data Structures, Software Engineering, Computer Systems and Programming, Analysis of Algorithms, Introduction to Low-level Programming, Formal Methods and Models

WORK EXPERIENCE

SAP NS2

May 2026 - Aug 2026

Language Model Intern

Herndon, VA

- Incoming Summer 2026

Lillup

Jan 2026 - Present

AI Software Engineer

Remote

- Built a local RAG pipeline that generates source-cited educational content using ChromaDB, LangChain, and GBNF-constrained generation via llama-cpp-python on quantized CPU models, resulting in consistently source-grounded output and reduced manual content curation effort.
- Develop a real-time text-to-speech pipeline that streams local LLM output with sentence buffering and TTS synthesis, using asyncio concurrent tasks and back-pressure-controlled queues across multiple style modes, enabling seamless voice interactions.

George Mason University

Jan 2026 - Present

Teaching Assistant

Fairfax, VA

- Supported a 292-student CS 262 course by answering conceptual and programming questions on Piazza, which improved student satisfaction and helped maintain high assignment completion rates
- Explained low-level programming concepts clearly and concisely to help students understand course material and assignment requirements.

NSF Spatiotemporal Innovation Center, George Mason University

Feb 2024 - Present

Undergraduate Research Assistant

Fairfax, VA

- Benchmarked SLMs and SRLMs on real-world Linux system logs, achieving 96% accuracy with retrieval-augmented prompting to evaluate deployability for real-time digital twin monitoring systems.
- Achieved 91% accuracy on conflict-text classification by comparing BERT and GPT using hyperparameter tuning and prompting methods such as RAG, RAG-ICL, and zero-shot approaches.
- Optimized context-based location-extraction performance to 90% through evaluation of open-source LLMs using various prompting methods and aligned evaluation methods.
- Reduced manual GIS processing time by 90% through automated conflict-mapping workflows using a multi-agent system (MAS) integrated with QGIS, Geocoding API, and Streamlit.

PROJECTS

Athena | <https://devpost.com/software/demo-e2sd8z>

- Built a real-time edge AI inference pipeline on Raspberry Pi, orchestrating STT, LLM reasoning (GPT-5 family), and TTS via asynchronous streaming with backpressure-controlled queues for low-latency response generation.
- Engineered a hybrid inference architecture combining on-device deterministic logic with cloud-based LLM routing, reducing unnecessary model calls and optimizing compute efficiency for resource-constrained systems.
- Implemented multi-modal interaction workflows including voice, visual generation (images/maps/diagrams), and interruptible streaming responses with robust error handling and low-latency performance on constrained edge hardware.

SKILLS

- **Programming Languages:** Python, C, Java
- **Frameworks/Libraries:** LangChain, ChromaDB, FAISS, llama-cpp-python, pandas, NumPy, PyTorch, scikit-learn
- **Tools & Platforms:** Git, Linux, AWS (EC2, Bedrock), PostgreSQL, MongoDB, LM Studio, Ollama, Huggingface, OpenClaw

SELECTED PUBLICATIONS

- Masri, Y., Ma, E., Wang, Z., Rogers, J. and Yang, C..Benchmarking Small Language Models and Small Reasoning Language Models on System Log Severity Classification.*arXiv preprint arXiv:2601.07790*
- Masri, Y., Wang, Z., Srirenganathan Malarvizhi, A., Ahmed, S., Stover, T., Wong, D.W., Jiang, Y., Li, Y., Liu, Q., Bere, M. and Rothbart, D..Comparative Analysis of BERT and GPT for Classifying Crisis News with Sudan Conflict as an Example.*Algorithms*, 18(7), p.420